Sectoral Reallocation and the Firm Life Cycle

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Introduction: Motivation

- How did changes in firm dynamics affect the sectoral reallocation of economic activity since the 1980s/1990s?
- Conceptually, sector *i* expands relative to sector *j* if:
  1. Incumbent firms in *i* grow relative to incumbent firms in *j*
  2. The marginal entrepreneur enters in *i* relative to *j*

$\rightarrow$ ΔEmployment $=$ ΔNumber of firms $+$ ΔAverage firm size

	Manufacturing			Services		
	Empl.	Firms	Avg. firm size	Empl.	Firms	Avg. firm size
1980	18.73	0.27	70.52	33.19	1.95	17.00
2018	11.86	0.23	51.68	82.74	3.46	23.91
Change (%)	-45.72	-14.64	-31.08	91.34	57.23	34.11

Notes: Employment and firms in millions. Data are from the Census Bureau's Business Dynamics Statistics (BDS).

Introduction: Methodology

1. Document new fact about life-cycle establishment growth across sectors
   - Establishments entering smaller and staying smaller, especially in manufacturing
   - Exit and job creation rates remained mostly stable among recent cohorts
2. Develop and estimate novel *multi-sector* firm dynamics model (building on Hopenhayn, 1992) to study contributing structural forces:
   i. Fixed costs of production (De Ridder, 2024)
   ii. Fixed costs of entry (Gutiérrez et al., 2021; Kozeniauskas, 2024)
   iii. Persistence of firm-level productivity (Decker et al., 2020)
   iv. Dispersion of firm-level productivity (Barth et al., 2016; Decker et al., 2020)
3. What if only within-sector dynamics changed between 1994 and 2018?
   - Sector-level firm dynamics attenuated aggregate structural change
   - Broad shifts in final demand across sectors remain the dominant force

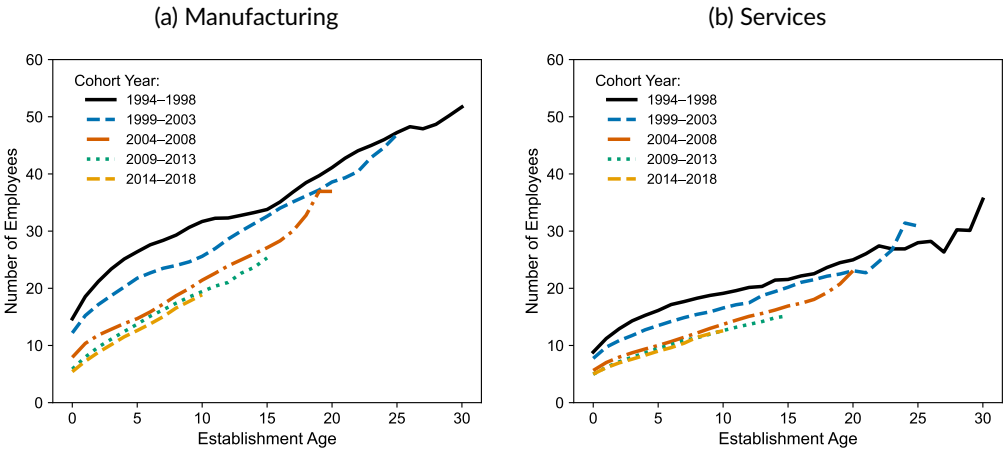
Roadmap

1. Motivating Evidence

2. Multi-Sector Firm Dynamics Model

3. Estimation and Decomposition

Trends in Average Establishment Size by Cohort



Source: Data are from the Business Employment Dynamics (BED) program of the BLS.

Regression

Multi-Sector Firm Dynamics Model

Endogenous mass of firms μ_i in each sector $i \in \{$Manufacturing (m), Services (s)$\}$

- Firms operate DRS production function; labor only input, no adjustment costs
- Idiosyncratic firm productivity evolves according to: $\ln(z') = \rho_i \ln(z) + \sigma_i \varepsilon$
- Firms choose labor input $l(z; p_i)$ and exit $X(z; p_i)$ policy to maximize

$$V(z; p_i) = \max_l [p_i z l^\alpha - wl - \phi_i] + \beta \max\{\mathrm{E}[V(z'; p_i)], 0\}$$

- Entrepreneurs pay sunk cost to draw initial z from G_i (stationary distribution)
  $\rightarrow$ Free entry $\implies \beta \int V(z; p_i)\, \mathrm{d}\, G_i(z) = \kappa_i$

Representative household consumes a bundle of goods across sectors

- $C = \left(\omega^{\frac{1}{\eta}} C_m^{\frac{\eta-1}{\eta}} + (1-\omega)^{\frac{1}{\eta}} C_s^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}$

Spillovers Across Sectors

Sector output shares are determined by sector weights ω_i and fundamentals θ_i

$$\frac{Y_i}{Y} = \omega_i \cdot \left(\frac{p_i(\theta_i)}{P} \right)^{-\eta}$$

- Consider an increase in fixed costs ϕ_i :
  - $\rightarrow$ More likely firms exit (selection effect: cutoff productivity $\bar{z}_i$ increases)
  - $\rightarrow$ $\downarrow$ Expected profits $\implies$ $\downarrow$ Mass of entrants M_i (acts as supply curve shifter)

    $\implies$ $\downarrow$ Sector output Y_i $\implies$ $\uparrow$ Sector price p_i
  - $\rightarrow$ Household substitutes away from sector i goods to sector j goods (or services)

Estimation Results

	Manufacturing					Services			
	ω	ρ_m	σ_m	ϕ_m	κ_m	ρ_s	σ_s	ϕ_s	κ_s
1994–1998	0.495	0.887	0.101	8.672	10.285	0.909	0.124	4.604	9.044
2014–2018	0.250	0.892	0.128	3.387	8.830	0.885	0.138	2.696	5.192

Targeted moments:

Moment	Level of Aggregation	Source
Average establishment size	Sector $\times$ Cohort $\times$ Age	BED
Establishment exit rate	Sector $\times$ Cohort $\times$ Age	BED
Job creation rate, expanding establishments	Sector $\times$ Cohort $\times$ Age	BED
Establishment entry rate	Sector $\times$ Year	BED
Value added share	Sector $\times$ Year	BEA-BLS

Counterfactual: Value Added Share of Manufacturing

	Data	Baseline	ρ_i	σ_i	ϕ_i	κ_i	ω
1994–1998	25.7	24.2	24.2	24.2	24.2	24.2	24.2
2014–2018	17.4	17.4	28.8	25.9	34.1	21.7	9.8
Difference (pp)	-8.3	-6.8	4.6	1.7	9.9	-2.5	-14.4
Percent change (%)	-32.3	-28.2	19.1	6.9	40.9	-10.3	-59.5
Explained (%)	100.0	82.1	-55.8	-20.2	-119.1	30.1	173.3

Data on value added are from the Integrated Industry-level Production Accounts (KLEMS) produced jointly by the BEA and BLS. Difference row shows 2014–2018 minus 1994–1998, in percentage points. Percent change row shows the percent change between 2014–2018 and 1994–1998, in percent. Explained row shows the respective counterfactual scenario divided by the Baseline, in percent.

Conclusion

Takeaways:

- In manufacturing sector, establishments entering smaller and staying smaller
  - Trend present in services sector, but to a lesser extent
  - → I argue this is an important moment to discipline theories of structural change
- Accounting for within-sector firm dynamics *dampens* structural change
  - → Absent these, fall in manufacturing value added share would have *doubled*

Next steps:

1. Further unpack changes in life-cycle dynamics using administrative data
   - Investigate role of multinational enterprises (MNEs), building on work linking Census LBD with BEA investment surveys (Kamal et al., 2022)
2. Extend model to capture additional interactions between sectors
   - Elastic sectoral-labor supply (Rogerson, 1987; Hopenhayn and Rogerson, 1993)
   - Intermediate goods (Bridgman and Herrendorf, 2024; Gaggl et al., 2025)

Thank You!

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Appendix

Regression Evidence

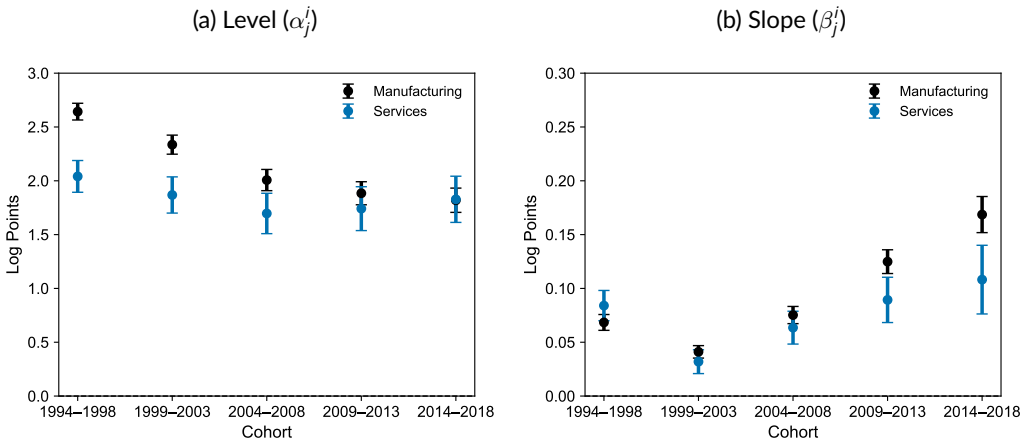
- Estimate sector-specific establishment dynamics by cohort (year of entry)
  - Let $i \equiv$ sector, $j \equiv$ cohort, $t \equiv$ year
  - Group cohorts into 5-year bins
- Parameterize as quadratic in establishment age

(1) $\ln(\text{Average Establishment Size})_{i,j,t} = \alpha_j^i + \beta_j^i \text{Age} + \gamma_j^i \text{Age}^2 + \delta_t + \varepsilon_{i,j,t}$

$\rightarrow$ Also estimate for Exit Rate and Job Creation Rate

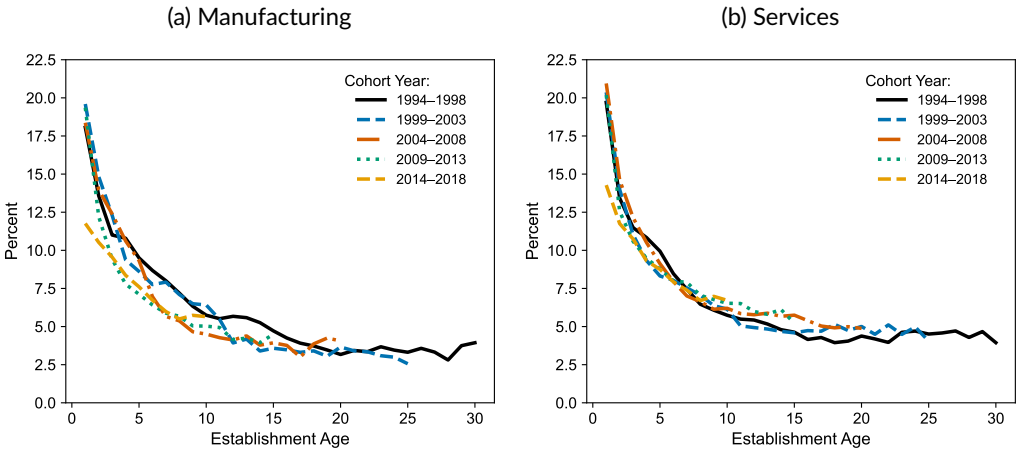
Back

Regression Evidence: Log(Average Establishment Size)



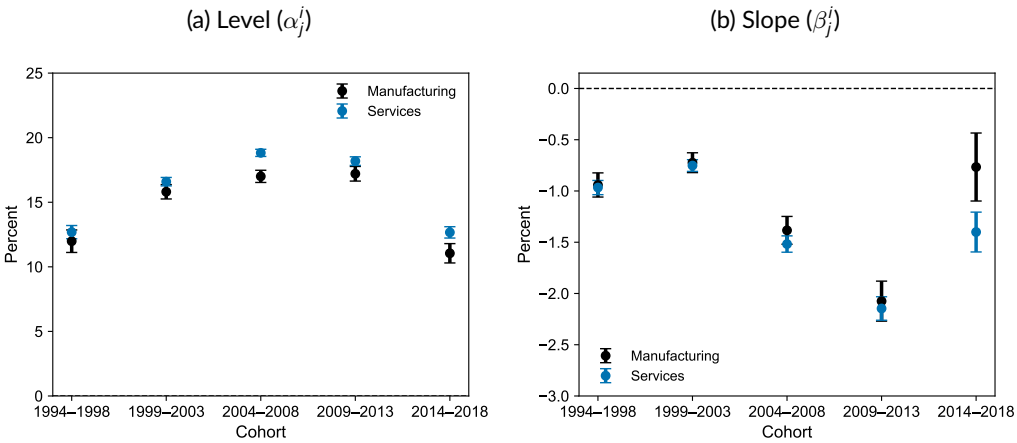
Notes: Data are from the Business Employment Dynamics (BED) program of the BLS. Average establishment size is defined as total employment divided by the total number of establishments.

Trends in Exit Rate by Cohort



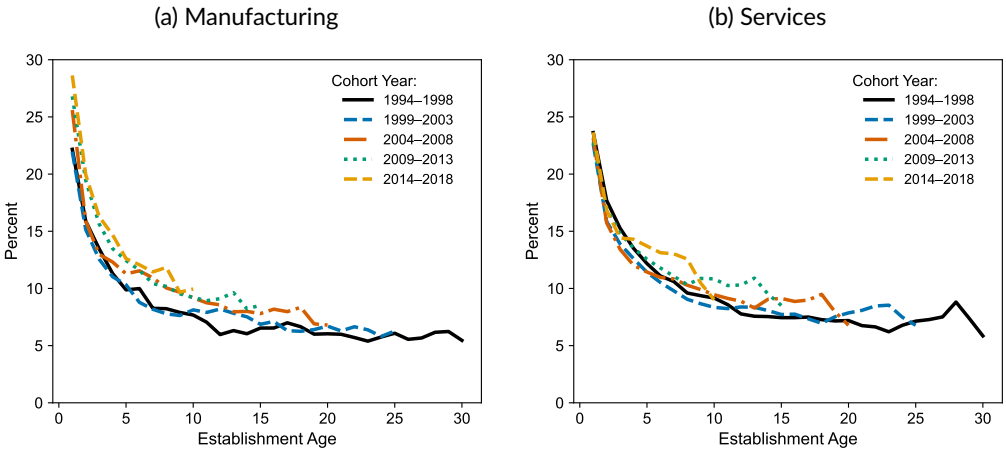
Source: Data are from the Business Employment Dynamics (BED) program of the BLS. The exit rate is defined as one minus the survival rate, where the survival rate is the number of surviving establishments in the cohort divided by this same value in the previous year.

Regression Evidence: Exit Rate



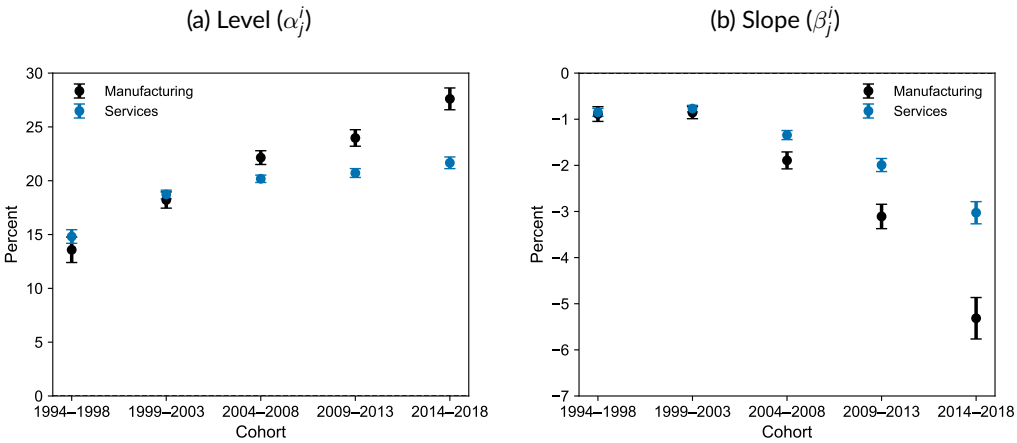
Notes: Data are from the Business Employment Dynamics (BED) program of the BLS. The exit rate is defined as one minus the survival rate, where the survival rate is the number of surviving establishments in the cohort divided by this same value in the previous year.

Trends in Job Creation Rate by Cohort



Source: Data are from the Business Employment Dynamics (BED) program of the BLS. The job creation rate is defined as the total number of jobs gained by expanding establishments divided by total employment.

Regression Evidence: Job Creation Rate



Notes: Data are from the Business Employment Dynamics (BED) program of the BLS. The job creation rate is defined as the total number of jobs gained by expanding establishments divided by total employment.

Classifying Sectors

Classification	NAICS Code	Sector Name
Manufacturing	31–33	Manufacturing
Services	51	Information
	52	Finance and insurance
	53	Real estate and rental and leasing
	54	Professional, scientific, and technical services
	55	Management of companies and enterprises
	56	Administrative and waste services
	61	Educational services
	62	Health care and social assistance
	71	Arts, entertainment, and recreation
	72	Accommodation and food services
	81	Other services (except public administration)

Notes: Sectoral classification follows groupings used by BLS: `https://www.bls.gov/iag/tgs/iag_index_naics.htm`